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Data Visualization Project Report

E-COMMERCE DASHBOARD REPORT

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Abstracts:

The report is about the e-commerce data analytics and data visualization project in which we use the MS Power BI to build the 3 interactive dashboards in Power BI. Our dataset has about 34,000 orders after removing the duplicate values from 2023 to 2025. We looked at sales, profits, product returns, customer types, and delivery times. We also started building a Tableau dashboard as part of this project. Main things we found: **Electronics makes up 56%** of all sales, the average delivery time is **4.8 days**, **84.9%** of orders are **profitable**, and the **return rate is 5.5%**. We give 6 simple recommendations to help the business improve.

Problem Statement:

The e-commerce company has \$6M in total sales, but management did not have a clear picture of what was happening behind the numbers. Here are the four main problems we wanted to solve:

- **High Returns:** 5.5% of orders are returned, costing the company money in refunds and restocking.
- **Unclear Profits:** Sales look good at \$6M, but which products and regions actually make money?
- **Delivery Link to Returns:** Nobody knew if late deliveries were causing customers to return items more.
- **No Targeted Strategy:** Decisions about marketing and inventory were based on guesswork, not data.

Objectives:

This project aimed to:

1. Find out why customers return products and how to reduce it.
2. Study **sales and profit** trends by month, region, and product category.
3. Check if late delivery leads to more returns.
4. Identify which customer types (age group, gender) spend the most.
5. Build **Power BI dashboards** and a **Tableau dashboard** for easy decision-making.

Dataset Overview:

We used an e-commerce dataset with about 34,000 orders. Here is a quick summary:

What	Detail
Total Orders	34,956 after duplicate values are removed
Time Period	2023, 2024, 2025
Regions	North, South, East, West, Central
Product Categories	7 — Electronics, Fashion, Home, Sports, Beauty, Toys, Grocery
Return Records	1,902 (5.5% of orders)
Payment Methods	6 — Credit Card, Debit, COD, UPI, PayPal, Wallet
Profit Status	Labeled as Profit or Loss per order

Key columns used in our analysis: order_id, customer_id, category, region, total_revenue, total_profit, delivery_days, return_reason, payment_method, and age_group.

Data Cleaning and Processing:

Before building the dashboards, we cleaned the data using **Power Query** inside Power BI:

- Removed duplicates — deleted repeated order IDs so each order was counted only once.
- Fixed missing values — blank return reasons were filled with 'No Return'; missing delivery days were filled using the average for that product category.
- Fixed data types — dates were converted to the correct format; revenue and profit were set as numbers.
- Created new columns — added: age_group, YearMonth, Return Flag (0 or 1), and Profit Status (Profit/Loss).
- Standardized text — removed extra spaces and fixed spelling differences in category and region names.

Checked outliers — kept valid edge cases, removed obvious data entry mistakes.

Methadology:

We followed these 5 steps:

1. **Data Collection** — loaded the CSV file into Power BI using the Get Data option.
2. **Data Cleaning** — used Power Query to fix, reshape, and prepare all data.
3. **Exploratory Analysis** — looked at patterns in sales, returns, delivery, and customer data.
4. **Dashboard Design** — built 3 Power BI dashboards with filters for Category, Year, and Region. Tableau dashboard is currently being built.
5. **Insights & Report** — turned dashboard findings into clear business recommendations.

We used DAX (Data Analysis Expressions) to create calculated measures such as:

- **Return Rate % = number of returned orders ÷ total orders**

- **Profit Margin % = total profit ÷ total revenue**
- **Avg Delivery Days = average of the delivery_days column**

Dashboard Building:

Dashboard 1 — Executive Overview (Power BI)

Shows the big picture. Key numbers at the top: **\$6M sales, 5.65M profit, 96.1% margin, 34K orders, 5.5% return rate**. Charts include: monthly sales trend, top 5 products, sales by category (**Electronics leads at 56%**), and a map showing sales by region. Orders by payment method shows **Credit Card** is the most used.

Dashboard 2 — Customer & Product Insights (Power BI)

Focuses on who buys what. A bar chart shows that **Adults (age 25-44)** spend the most, and **female customers** slightly outspend males. An area chart shows average order value has dropped from **\$188 in 2023 to \$152 in 2025**. A table breaks down sales by category and region — **Electronics is the biggest** in every region.

Dashboard 3 — Returns Analysis (Power BI)

Shows why and where products are returned. A donut chart shows 4 nearly equal return reasons: Not as Described (**25.76%**), Defective (**25.24%**), No Longer Needed (**24.45%**), and Other (**23.08%**). **South** has the **highest returns**. Average delivery time is 4-8 days but return rates go up when delivery takes 5-7 days. **84.9%** of orders are profitable.

Results and Key Findings:

#	Finding	What It Means
F1	Electronics = 56% of Sales	It is the biggest revenue driver — \$3.26M out of \$6M total.
F2	Returns Are Split Evenly	No single reason causes most returns. All 4 reasons are ~25% each.
F3	4.8 Days Avg Delivery	Stable, but returns spike when delivery takes 5-7 days.
F4	South Has Most Returns	South region: 0.43K returns. East & North: 0.41K each.
F5	84.9% Orders Are Profitable	Good health, but 15.1% still make a loss — needs review.
F6	Adults (25-44) Buy the Most	Over \$1M in sales from this age group. Females spend slightly more.
F7	Avg Order Value Dropping	Fell from \$188 (2023) to \$152 (2025) — a \$36 per order drop.

Recommendations:

Improve Product Listings

25.76% of returns happen because the product was 'Not as Described'. Better photos, accurate descriptions, and size guides will directly reduce these returns.

Focus on South Region

South has the highest return rate. The company should investigate whether local suppliers or courier services are the cause and run regional promotions to rebuild trust.

Protect Electronics Revenue

Electronics is 56% of revenue. Keep stock levels high, add quality checks before shipping, and offer warranty support to protect this top category.

Loyalty Program for Adults (25-44)

Adults are the top buyers. A loyalty card, reward points, or personalized offers for this group will increase repeat purchases and average order value.

Improve Delivery in the 5-7 Day Window

Returns spike when delivery takes 5-7 days. Better courier SLAs, proactive tracking updates, and express shipping for electronics can help.

Review Loss-Making Orders

15.1% of orders lose money. These are likely caused by heavy discounts. Setting minimum margins on discounts will turn many of these into profit.

Tools & Technologies:

- Power BI Desktop, Power Query (M), DAX
- Tableau

Challenges:

Inconsistent Raw Data

The data had mixed date formats, spelling errors in category names, and missing values. We fixed all of this using Power Query before building the dashboards.

Writing DAX Formulas

Measures like Return Rate % had to respond correctly when users clicked filters. We tested each formula step-by-step using Power BI's formula bar and checked results against manual counts.

Slow Dashboard Performance

The dashboard was slow at first because of the large data size. We removed columns we did not need and used summary tables for heavy calculations, which made it faster.

Conclusion:

This project showed how data visualization tools like Power BI can turn raw sales data into useful business insights. We built 3 working Power BI dashboards and a Tableau dashboard is currently being developed.

Our most important findings: Electronics dominates sales at 56%, return reasons are spread evenly, delivery time of 4.8 days correlates with returns at 5-7 days, and 84.9% of orders are profitable. We gave 6 clear recommendations that the business can act on immediately.

Future work could include: connecting to a live data feed, using machine learning to predict which orders will be returned before they ship, and completing the Tableau dashboard for comparison with Power BI.